

Mark schemes

Q1.

(a) $\int x^2 e^{-x} dx = -x^2 e^{-x} - \int -2x e^{-x} dx$
 $kx^2 e^{-x} - \int kx e^{-x} (dx) \text{ for } k = \pm 1$

M1
A1

$= -x^2 e^{-x} + 2\{-x e^{-x} - \int -e^{-x} dx\}$
 $\int x e^{-x} dx = \lambda x e^{-x} - \int \lambda e^{-x} (dx) \text{ for } \lambda = \pm 1 \text{ in 2nd}$
application of integration by parts

m1

$= -x^2 e^{-x} - 2x e^{-x} - 2e^{-x} (+c)$
Condone absence of + c

A1

4

(b) $I = \int_0^{\infty} x^2 e^{-x} dx = \lim_{a \rightarrow \infty} \int_0^a x^2 e^{-x} dx$
F(a) - F(0) with an indication of limit 'a → ∞' and F(x)
containing at least one $x^n e^{-x}$, $n > 0$ term

M1

$\lim_{a \rightarrow \infty} \{-a^2 e^{-a} - 2a e^{-a} - 2e^{-a}\} - [-2]$

$\lim_{a \rightarrow \infty} a^k e^{-a} = 0, \quad (k > 0)$
For general statement or specific
statement for either $k = 1$ or $k = 2$

E1

$\int_0^{\infty} x^2 e^{-x} dx = 2$
 CSO

A1

2

[6]

Q2.

- (a) The interval of integration is infinite

OE

E1

1

(b) (i) $x = \frac{1}{y} \Rightarrow dx = -y^{-2} dy$

$$\int \frac{\ln x^2}{x^3} dx \Rightarrow \int (y^3 \ln y^{-2}) (-y^{-2}) dy$$

M1

$$= \int -y \ln y dy = \int 2y \ln y dy$$

CSO AG

A1

2

(ii) $\int 2y \ln y dy = y^2 \ln y - \int y^2 \left(\frac{1}{y} \right) dy$

... = $ky^2 \ln y \pm \int f(y) dy$ with $f(y)$ not involving the 'original' $\ln y$

M1A1

$$\dots = y^2 \ln y - \frac{1}{2} y^2 + c$$

Condone absence of '+ c'

A1

$$\int_0^1 2y \ln y dy = \lim_{a \rightarrow 0} \int_a^1 2y \ln y dy$$

$$= \left(0 - \frac{1}{2} \right) - \lim_{a \rightarrow 0} \left[a^2 \ln a - \frac{a^2}{2} \right]$$

M1

$$= -\frac{1}{2} \text{ since } \lim_{a \rightarrow 0} a^2 \ln a = 0$$

CSO Must see clear indication that cand has correctly considered

$$\lim_{a \rightarrow 0} a^k \ln a = 0$$

A1

5

(iii) So $\int_1^{\infty} \frac{\ln x^2}{x^3} dx = \frac{1}{2}$

ft on minus c's value as answer to (b)(ii)

B1F

1

[9]

Q3.

- (a) The interval of integration is infinite
OE

E1

1

(b) $\int 4xe^{-4x} dx = -xe^{-4x} - \int -e^{-4x} dx$
 $kxe^{-4x} - \int ke^{-4x} dx$ for non-zero k

M1A1

$$= -xe^{-4x} - \frac{1}{4} e^{-4x} \{+c\}$$

Condone absence of + c

A1F

3

(c) $I = \int_1^{\infty} 4xe^{-4x} dx = \lim_{a \rightarrow \infty} \int_1^a 4xe^{-4x} dx$

$$\lim_{a \rightarrow \infty} \left\{ -ae^{-4a} - \frac{1}{4} e^{-4a} \right\} - \left[-\frac{5}{4} e^{-4} \right]$$

F(a) - F(1) with an indication of limit 'a → ∞'

M1

$$\lim_{a \rightarrow \infty} ae^{-4a} = 0$$

For statement with limit/ limiting process shown

M1

$$I = \frac{5}{4} e^{-4}$$

CSO

A1

3

[7]



Q4.

$$(a) \int \ln x \, dx = x \ln x - \int x \left(\frac{1}{x} \right) dx$$

Integration by parts

M1

$$= x \ln x - x + C$$

CSO

AG

A1

2

$$(b) \int_0^1 \ln x \, dx = \lim_{a \rightarrow 0} \int_a^1 \ln x \, dx$$

OE

M1

$$= \lim_{a \rightarrow 0} \{0 - 1 - [a \ln a - a]\}$$

F(1) - F(a) OE

M1

$$\text{But } \lim_{a \rightarrow 0} a \ln a = 0$$

Accept a general form eg

$$\lim_{a \rightarrow 0} a^k \ln a = 0$$

E1

$$\text{So } \int_0^1 \ln x \, dx = -1$$

A1

4

[6]

Q5.

(a) The interval of integration is infinite

OE

E1

1

$$(b) \int x e^{-3x} \, dx = -\frac{1}{3} x e^{-3x} - \int -\frac{1}{3} e^{-3x} \, dx$$

Reasonable attempt at parts

M1A1



$$= -\frac{1}{3} x e^{-3x} - \frac{1}{9} e^{-3x} \{+c\}$$

Condone absence of + c

A1ft

3

(c)
$$I = \int_1^{\infty} x e^{-3x} dx = \lim_{a \rightarrow \infty} \int_1^a x e^{-3x} dx$$

$$\lim_{a \rightarrow \infty} \left\{ -\frac{1}{3} a e^{-3a} - \frac{1}{9} e^{-3a} \right\} - \left[-\frac{4}{9} e^{-3} \right]$$

$F(a) - F(1)$ with an indication of limit ' $a \rightarrow \infty$ '

M1

$$\lim_{a \rightarrow \infty} a e^{-3a} = 0$$

For statement with limit/limiting process shown

M1

$$I = \frac{4}{9} e^{-3}$$

A1

3

[7]

Q6.

(a)
$$\int x^3 \ln x dx = \frac{x^4}{4} \ln x - \int \frac{x^4}{4} \left(\frac{1}{x} \right) dx$$

... = $\frac{x^4}{4} \ln x \pm \frac{x^4}{16}$, with $f(x)$ not involving the 'original' $\ln x$

M1A1

$$\dots = \frac{x^4}{4} \ln x - \frac{x^4}{16} + c$$

Condone absence of '+ c

A1

3

(b) Integrand is not defined at $x = 0$

OE

E1

1



(c) $\int_0^e x^3 \ln x \, dx = \left\{ \lim_{a \rightarrow 0} \int_a^e x^3 \ln x \, dx \right\}$
 $= \frac{3e^4}{16} - \lim_{a \rightarrow 0} \left[\frac{a^4}{4} \ln a - \frac{a^4}{16} \right]$
 $F(e) - F(a)$

M1

But $\lim_{a \rightarrow 0} a^4 \ln a = 0$

Accept a general form eg $\lim_{x \rightarrow 0} x^4 \ln x = 0$

B1

So $\int_0^e x^{-3} \ln x \, dx$ exists and $= \frac{3e^4}{16}$
 CSO

A1

3

[7]

Q7.

- (a) Integrand is not defined at $x = 0$

OE

E1

1

(b) $\int x^{\frac{1}{2}} \ln x \, dx = 2x^{\frac{1}{2}} \ln x - \int 2x^{\frac{1}{2}} \left(\frac{1}{x} \right) dx$

M1

$\dots = kx^{\frac{1}{2}} \ln x \pm \int f(x),$ with $f(x)$ not involving
 the 'original' $\ln x$

A1

$\dots = 2x^{\frac{1}{2}} \ln x - 4x^{\frac{1}{2}} (+c)$

Condone absence of '+c'

A1

3

(c) $\int_0^e \frac{\ln x}{\sqrt{x}} \, dx = \lim_{a \rightarrow 0} \int_a^e \frac{\ln x}{\sqrt{x}} \, dx$

M1

$$= -2e^{\frac{1}{2}} - \lim_{a \rightarrow 0} \left[2a^{\frac{1}{2}} \ln a - 4a^{\frac{1}{2}} \right]$$

$F(b) - F(a)$

M1

But $\lim_{a \rightarrow 0} a^{\frac{1}{2}} \ln a = 0$

Accept a general form e.g. $\lim_{x \rightarrow 0} x^k \ln x = 0$

B1

So $\int_0^e \frac{\ln x}{\sqrt{x}} dx$ exists and $= 2e^{\frac{1}{2}}$

A1

4

[8]

Q8.

(a) 0

B1

1

(b) $u = xe^{-x} + 1 \Rightarrow du = (e^{-x} - xe^{-x}) dx$

Attempts to find du

M1

$$\int \frac{e^{-x}(1-x)}{xe^{-x}+1} dx = \int \frac{1}{u} du = \ln u + c = \ln(xe^{-x}+1) + c$$

Condone missing c

A1

2

(c) $\int \frac{1-x}{x+e^x} dx = \int \frac{e^{-x}(1-x)}{xe^{-x}+1} dx$

B1

$$\int_1^\infty \frac{1-x}{x+e^x} dx = \lim_{a \rightarrow \infty} [\ln(xe^{-x}+1)]_1^a$$

$$= \lim_{a \rightarrow \infty} \{\ln(ae^{-a}+1)\} - \ln(e^{-1}+1)$$

For using part (b) and $F(B) - F(A)$

M1

$$= \ln \left\{ \lim_{a \rightarrow \infty} (ae^{-a} + 1) \right\} - \ln(e^{-1} + 1)$$

$$= \ln 1 - \ln(e^{-1} + 1) = -\ln(e^{-1} + 1)$$

For using limiting process

M1

A1

4

[7]

Q9.

$$(a) \int xe^{-2x} dx = -\frac{1}{2} xe^{-2x} - \int -\frac{1}{2} e^{-2x} dx$$

M1

Reasonable attempt at parts

A1

$$= -\frac{1}{2} xe^{-2x} - \frac{1}{4} e^{-2x} \{+c\}$$

Condone absence of + c

A1ft

$$\int_0^a xe^{-2x} dx = -\frac{1}{2} ae^{-2a} - \frac{1}{4} e^{-2a} - (0 - \frac{1}{4})$$

 $F(a) - F(0)$

M1

$$\frac{1}{4} - \frac{1}{2} ae^{-2a} - \frac{1}{4} e^{-2a}$$

A1

5

lim

$$(b) \lim_{a \rightarrow \infty} ae^{-2a} = 0$$

B1

1

$$(c) \int_0^{\infty} xe^{-2x} dx =$$



$$\lim_{a \rightarrow \infty} \left\{ \frac{1}{4} - \frac{1}{2} a e^{-2a} - \frac{1}{4} e^{-2a} \right\}$$

If this line oe is missing then 0/2

M1

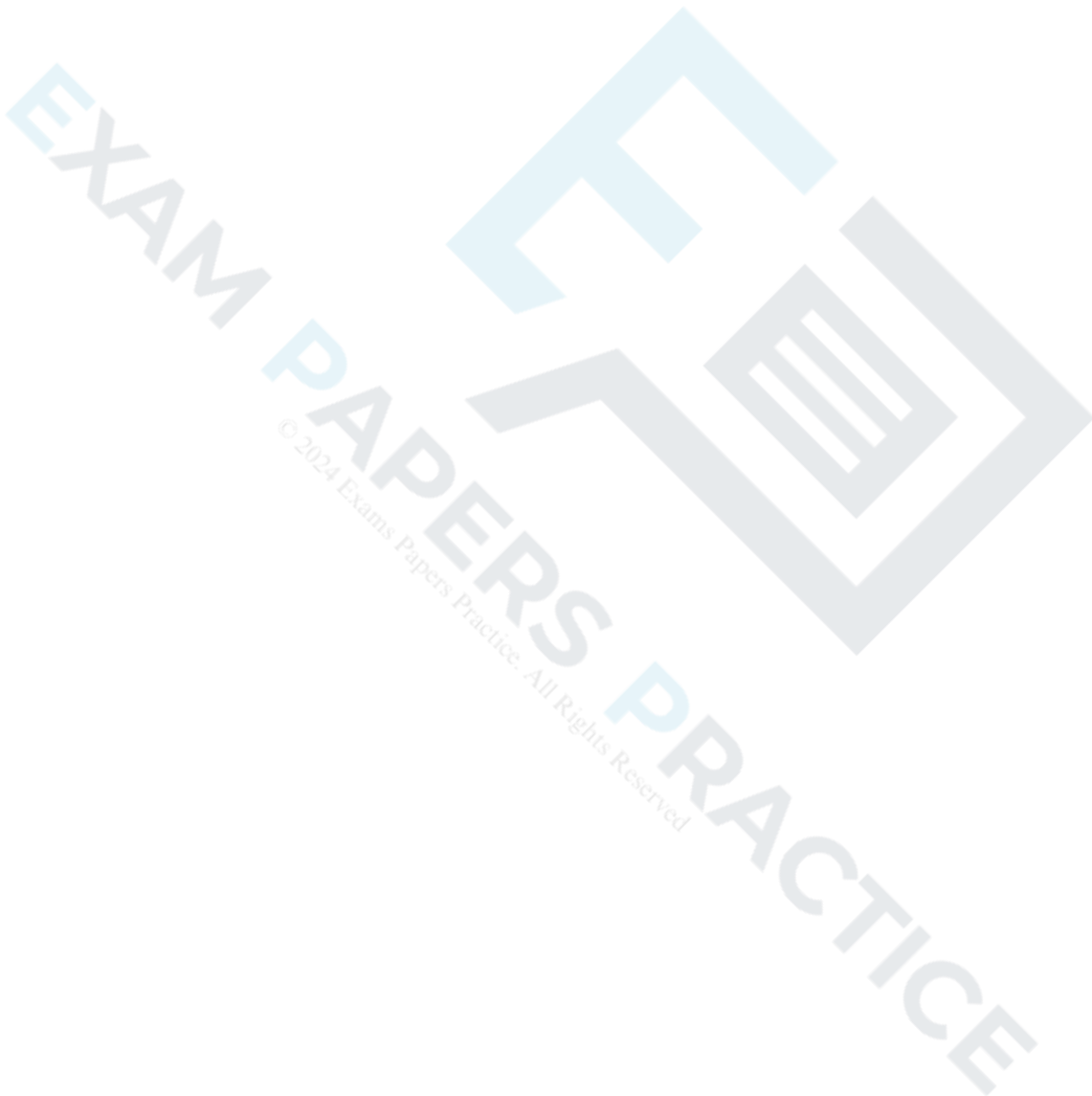
$$= \frac{1}{4} - 0 - 0 = \frac{1}{4}$$

*On candidate's "1/4" in part (a).
B1 must have been earned*

A1ft

2

[8]



Examiner reports

Q1.

Part (a) was, as expected, generally answered well, with students clearly showing the method of integration by parts. In part (b), a higher proportion of students than in previous papers replaced the upper limit with a constant (for example, a) indicated that $a \rightarrow \infty$, and then found $F(a) - F(0)$. However, a significant number of students were not explicit in indicating that the limit as $a \rightarrow \infty$, $a^k e^{-a} = 0$ and hence they lost the final two marks. This is illustrated by the common error which was just to state that as $a \rightarrow \infty$, $e^{-a} \rightarrow 0$ so given integral = 2.

Q2.

In part (a), a significant number of candidates failed to give the correct reason for why the integral was improper, with most giving the reason that the integrand was undefined for the upper limit rather than that the interval of integration was infinite. In part (b)(i), a large number of candidates failed to use the substitution correctly. Part (b)(ii) was generally answered well with most candidates integrating by parts correctly to score 3 of the 5 marks available, but some candidates did not gain the final two marks as examiners did not see the lower limit replaced by, for example, a and the consideration of the limiting process as $a \rightarrow 0$. Part (b)(iii) was answered incorrectly by the majority of candidates, who usually either stated the same value or the reciprocal value of their answer to part (b)(ii).

Q3.

Improper integrals and limiting processes continue to cause problems for a significant minority of candidates. Part (a) was generally not answered well with too many candidates making a statement which they then contradicted in part (c). As highlighted in the MFP3 examiners' reports for January 2008 and January 2010, an integral, with ooas a limit, is improper because the interval of integration is infinite.

Most candidates correctly applied integration by parts to score the three marks available in part (b).

In part (c) examiners expected to see the infinite upper limit replaced by, for example, a , the integration carried out and then consideration of the limiting process as $a \rightarrow \infty$.

It was not uncommon to see in solutions the statement 'as $a \rightarrow \infty$, $e^{-4a} \rightarrow 0$ so $e^{-4a}(-a - 1/4) \rightarrow 0$ ' without seemingly any particular analysis of $\lim_{a \rightarrow \infty} a e^{-4a}$. This lack of analysis was penalised.

Q4.

Part (a) was, as expected, very well answered with candidates clearly showing the method of integration by parts. In part (b), a smaller proportion of candidates than in June 2008 showed the limiting process fully. For full marks candidates were also expected to pay particular attention to the value of the limit of $(a \ln a)$ as a tended to 0.

Q5.

Part (a) was generally not well answered with a significant minority either not attempting it or making a statement which they then contradicted in part (c). The method of integration

by parts was understood with the great majority obtaining the correct answer to part (b). Although there continues to be an improvement in candidates' solutions to the evaluation of an improper integral, there were still a significant minority who made no attempt to show the limiting process used.

Q6.

Most candidates applied integration by parts accurately to obtain the correct answer for the integral of $x^3 \ln x$. Although many candidates gave a correct explanation in part (b) for why the integral was improper, there were others whose incorrect explanations centred on the limit e or the interval of integration being infinite. For full marks in part (c), candidates were expected to pay particular attention to the value of the limit of, for example, $a^4 \ln a$ as a tended to 0.

Q7.

A significant minority did not attempt part (a), or used the limit e rather than 0 in their explanations of why the definite integral was improper. Using integration by parts in part (b) proved to be more of a problem for candidates than expected. Although a higher proportion of candidates showed the limiting process than in previous sessions, this topic still remains a section of the specification which is generally not fully understood.

Q8.

The standard result required in part (a) was known by most candidates. In part (b), a minority of candidates failed to use the given substitution and a significant minority of others left their final answer in terms of u . In part (c) it was not always clear to the examiners that a candidate had fully seen the relationship between the integrands in parts (b) and (c). Some candidates clearly showed the relationship by multiplying both the numerator and the denominator of the integrand in part (c) by e^{-x} . Sometimes poor use of brackets led to the wrong answer '2' presumably from $-\ln e^{-1} + 1$. However it was pleasing to see the increased number of candidates who correctly used the limiting process this year.

Q9.

Most candidates applied integration by parts correctly in part (a) but it was disappointing to see a significant minority not attempting to substitute the given limits. The standard result required in part (b) was well known. In the final part, although many showed the limiting process used, a significant number of candidates just replaced x by ∞ without comment.